

InvisiGuard - Intelligent System for Predicting Cyber Threats Real Time

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Declaration

I declared that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Signature

Date

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

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Abstract

Traditional cybersecurity systems mainly use signature-based detection methods. These methods work well for finding known threats but often miss new, evolving, and previously unseen attacks. As cyber threats become more complex, there is a greater need for smart and proactive security solutions that can anticipate attacks before they cause serious damage. This research aims to create a Real Time Cyber Threats Predicting System that employs machine learning to predict both known and zero-day cyberattacks. The system is structured as a multi-component framework with four main parts: Predicting Known Attacks Using System Call Patterns, Predicting Known Attacks Using User Behavioral Patterns, Predicting Zero-Day Attacks Using Network Traffic Patterns, and Predicting Known Attacks Using Endpoint Log Patterns. Together, these components form a layered cybersecurity solution that can improve threat detection and prediction.

This study will focus on the Predicting Zero-Day Attacks Using Network Traffic Patterns part. Zero-day attacks are particularly difficult to handle because they take advantage of unknown vulnerabilities, making it hard to detect using traditional rule-based or signature-based methods. To overcome this challenge, the proposed system uses unsupervised machine learning techniques to analyze network traffic and spot anomalies that might indicate malicious activity. By learning what normal network traffic looks like, the model can identify unusual changes and flag them as possible zero-day threats. This helps the system detect suspicious patterns in real time, even without prior knowledge of the attack.

This approach improves the effectiveness of the Real Time Cyber Threats Predicting System by adding an intelligent way to identify zero-day threats early. By detecting anomalies in network traffic patterns, the system enhances proactive cybersecurity defenses, reduces response time, and lessens the potential damage from unknown attacks. As a result, this research contributes to building a more adaptable, predictive, and intelligence-driven cybersecurity framework that strengthens organizational readiness against emerging cyber threats.

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1.Introduction

In recent years, the number and complexity of cyberattacks have increased significantly due to our growing reliance on digital technologies, cloud platforms, online services and interconnected systems across different industries. Organizations in finance, healthcare, education, government and telecommunications depend heavily on networked systems for their daily operations. As this reliance grows, cybercriminals keep developing more sophisticated attack methods that can overcome standard security measures. Zero-day attacks have become one of the most dangerous and challenging types of cyberattacks. A zero-day attack happens when attackers take advantage of software, hardware or system vulnerability before the vendor or developer is aware and can release a patch or fix. Since vulnerability is unknown, there is no signature, rule or predefined pattern for traditional intrusion detection systems (IDS) or antivirus tools to identify it effectively. This makes zero-day attacks particularly dangerous, as they can remain hidden while causing severe damage, including unauthorized access, system disruptions, data theft, financial loss and reputational harm.

Several major incidents in recent years have underscored the seriousness of zero-day threats. In 2021, the Log4Shell (CVE-2021-44228) vulnerability shocked the cybersecurity community because it allowed attackers to execute malicious code on millions of vulnerable servers globally. The flaw affected many large organizations and technology providers, including Amazon, Microsoft and Cloudflare. Similarly, in 2023, the MOVEit Transfer zero-day vulnerability led to the breach of hundreds of organizations and exposed sensitive data belonging to millions of people. These cases clearly show how quickly zero-day vulnerabilities can spread and how damaging they can be when traditional detection tools fail. These real-world examples indicate that relying solely on signature-based detection is no longer adequate in today's cybersecurity landscape. Signature-based systems can detect known attacks effectively, but they are reactive. They rely on recognized attack patterns, which means they cannot guard against new, evolving, or unseen threats. As attackers continually change their tactics, security systems need to be smarter, more adaptive and proactive.

To tackle these challenges, organizations require more than traditional intrusion detection methods or simple machine learning models based only on supervised learning. While supervised machine learning can effectively classify known attack types, it relies heavily on labeled datasets. This creates a significant limitation for predicting zero-day attacks since they are inherently unknown and generally lack labeled training examples. A better solution is needed, one that can spot abnormal or suspicious behavior even when no prior attack signature is available. One promising approach is using anomaly detection based on network traffic analysis. Network traffic provides useful insights into how systems interact, how data flows between devices and whether unusual patterns are present. Zero-day attacks often generate abnormal communication behavior, so reviewing traffic patterns can help identify suspicious activity before major damage occurs. By

understanding what normal network behavior looks like, machine learning models can detect deviations that may signal an unknown attack. [1]

This report discusses the Real-Time Intelligent System for Predicting Cyber Threats, with a specific focus on predicting zero-day attacks through network traffic patterns. This component aims to identify potential zero-day attacks by analyzing network traffic behavior using anomaly detection techniques. Instead of relying on known signatures, it aims to spot unusual traffic patterns that could indicate previously unseen malicious activity. This makes the method more proactive and better suited to today's cybersecurity needs. Furthermore, this subcomponent is vital in the larger system because network traffic is among the earliest and most informative indicators of cyber threats. Suspicious traffic flows, unusual protocol usage, unexpected communication frequency or irregular packet behavior can often signal an upcoming attack before it fully impacts the target system. Therefore, by emphasizing network traffic analysis, this research helps build a stronger and smarter framework for predicting cyber threats.

Overall, the rising frequency of zero-day attacks and the shortcomings of traditional detection systems highlight the pressing need for more advanced predictive methods. This report explores how network traffic-based anomaly detection can enhance zero-day attack prediction and strengthen real-time cybersecurity defenses.

1.1 Background and Literature

Cybersecurity threats are becoming more sophisticated. Traditional security systems that rely on predefined attack signatures struggle to detect new and unknown threats, particularly zero-day attacks. A zero-day attack happens when a vulnerability is exploited before developers or security vendors are aware of it and release a security patch. Since these attacks do not match any known signatures, signature-based intrusion detection systems often fail to identify them effectively. This challenge has created a strong demand for smarter and more flexible detection methods. To tackle this issue, researchers have increasingly explored machine learning techniques in cybersecurity. Machine learning allows systems to learn patterns from data instead of depending solely on predefined rules. In cyber threat detection, machine learning models can be trained to recognize what normal behavior looks like in network traffic, user activity, or system logs. Once the model learns normal behavior, it can spot deviations or abnormal patterns that may signal malicious activity. This makes machine learning particularly valuable for detecting previously unseen threats.

Many past studies have focused on using machine learning models to identify attacks by analyzing specific types of data, such as network traffic, system calls or user behavior. These studies have shown that machine learning can improve detection over traditional systems. However, much of the existing research is limited in scope. Many researchers focus solely on one area of cyber threat analysis without considering the broader context of real-time cyber threat prediction. In the larger research project, the system is divided into four main areas: system calls, user behaviors, network

traffic patterns, and endpoint logs. Existing studies have not thoroughly covered the complete range of predictions across these sources, which creates an important research gap. This broader limitation is one reason why the main research proposes a real-time intelligent system for predicting cyber threats. This system aims to examine cyber threats from multiple perspectives rather than relying on one source of data. My component specifically focuses on predicting zero-day attacks using network traffic patterns, a crucial area since network traffic often shows early signs of suspicious or unusual behavior. Network traffic contains valuable information such as communication patterns, protocol usage, packet flow characteristics and unusual transmission activity. These patterns can uncover hidden threats, even when the specific attack type is unknown.

In the realm of zero-day attack prediction, many studies have used machine learning techniques, but many of them depend on supervised learning. Supervised learning methods require labeled data during training, meaning each training sample must be classified as normal or malicious. This creates a major challenge for zero-day attack prediction because zero-day attacks are, by definition, previously unknown. As labeled examples of unknown attacks are rarely available, supervised methods are not the most effective solution for this problem. When models rely too heavily on labeled data, their accuracy and practical usefulness can drop when new threats differ from the training data. Due to this limitation, this component uses unsupervised learning techniques instead of supervised learning. Unsupervised learning does not depend on labels and identifies patterns, structures, and irregularities directly from the data. This makes it more suitable for detecting zero-day attacks, where the attack behavior may not have been seen before. By learning what normal network behavior looks like, unsupervised models can find unusual traffic that may indicate a zero-day threat. This allows the detection system to be more adaptable and proactive.

One of the main techniques used in this component is the Autoencoder, a type of neural network designed to reconstruct input data. In this context, the Autoencoder learns to reconstruct normal network traffic patterns. When it receives traffic like normal behavior, the reconstruction error stays low. However, when it gets unusual or anomalous traffic, the reconstruction error increases. This makes the Autoencoder effective for anomaly detection in cybersecurity. The model architecture used in this study is a fully connected neural network with encoder and decoder layers arranged as follows: input layer, dense layer with 64 neurons, dense layer with 32 neurons, bottleneck layer with 16 neurons, dense layer with 32 neurons, dense layer with 64 neurons and output layer. The anomaly decision is based on Mean Squared Error reconstruction loss. In addition to the Autoencoder, this component also employs Isolation Forest, another unsupervised anomaly detection algorithm. Unlike Autoencoders, Isolation Forest does not focus on reconstruction. Instead, it finds anomalies by isolating unusual data points through random partitioning. Since anomalies are generally rare and different from most data, they tend to be separated more quickly in the tree structure. This makes Isolation Forest efficient for handling large, high-dimensional datasets like network traffic logs. It is especially useful when the dataset contains many features and when the exact nature of abnormal behavior is unclear. The Isolation Forest model in this research uses 300 trees, a contamination rate of 0.01, and a random state of 42.

While both Autoencoder and Isolation Forest are useful on their own, each has limitations. Autoencoders can learn complex patterns but may still produce false positives since some abnormal samples can be reconstructed quite well. Isolation Forest is efficient for isolating anomalies but might struggle to capture complex structures in high-dimensional data. Because of these limitations, this research combines both techniques to create a hybrid model. The hybrid approach uses the Autoencoder reconstruction error and the Isolation Forest anomaly score, combines them into a feature vector, and applies a Logistic Regression meta-classifier to make the final prediction. This allows the system to leverage the strengths of both models to enhance the reliability of anomaly detection. Another issue found in previous studies is the use of outdated datasets. Some research has applied machine learning and anomaly detection techniques using old or unrealistic datasets that do not reflect current network conditions. As cyberattacks continue to evolve, training models on outdated datasets reduces the system's ability to accurately identify modern attack behavior. Some datasets also lack enough diversity in traffic patterns and attack scenarios, limiting their effectiveness in zero-day prediction.

To tackle this issue, this component uses the HIKARI-2021 dataset, which is more suitable for modern network intrusion detection research. HIKARI-2021 contains realistic network traffic and includes both normal and malicious activity, such as encrypted communications, making it more representative of current cybersecurity environments. This dataset is particularly useful for this research because it contains various traffic behaviors and attack scenarios. For Autoencoders, it offers rich normal traffic patterns that help the model learn typical behavior and detect deviations through reconstruction errors. For Isolation Forest, the diversity of traffic features and attack scenarios enhances the model's ability to identify unusual or rare events. Therefore, HIKARI-2021 is well-suited for hybrid machine learning methods aimed at predicting zero-day attacks. Another significant gap in literature exists. Some studies use unsupervised techniques and appropriate datasets, but they still focus only on general anomaly detection or intrusion detection instead of directly tackling zero-day attack prediction. Predicting zero-day attacks is more challenging because it involves identifying behaviors linked to vulnerabilities that are unknown at the time of detection. This requires not only anomaly detection but also a stronger predictive approach that can highlight suspicious network patterns before a threat is formally acknowledged. This study aims to meet that challenge by building a more effective hybrid system specifically for zero-day prediction based on network traffic behavior.

In summary, machine learning offers a strong alternative to traditional signature-based methods for detecting zero-day attacks. By learning normal network behavior and recognizing unusual deviations, machine learning systems can detect new and unknown threats more effectively. This component builds on that idea by combining Autoencoder and Isolation Forest models with a modern dataset and a hybrid architecture. As a result, it supports a more adaptable and proactive method for predicting zero-day attacks in real-time cybersecurity environments. [2]

1.2 Research gap

Zero-day attacks are some of the hardest cybersecurity threats to spot because they take advantage of unknown vulnerabilities. Traditional network security systems mainly use signature-based detection methods. These methods only work when the attack pattern has already been identified and cataloged. Because of this, these systems are effective against known threats but struggle to detect new attack behaviors. Since zero-day attacks lack existing signatures, they can evade many standard intrusion detection systems before security teams notice them. This creates a major weakness in current cybersecurity defenses. To address this issue, researchers have investigated anomaly-based intrusion detection systems (IDS). Unlike signature-based systems, anomaly-based IDS models learn what normal network behavior looks like and then spot deviations that might signal suspicious activity. This makes them better at detecting unknown threats. However, while past studies have improved anomaly detection, significant research gaps still exist in zero-day attack prediction using network traffic patterns.

One major limitation in earlier research is that some studies focused on general intrusion detection rather than zero-day attack prediction. These studies might successfully find known attacks and some unusual traffic patterns, but they do not specifically tackle the challenge of predicting new, unseen attacks. Identifying a general anomaly is not always the same as predicting a zero-day attack. Zero-day situations require a proactive approach that can spot subtle, suspicious deviations early, before the attack is officially recognized.

Another limitation is the reliance on outdated datasets in previous studies. Network environments change constantly, and attackers often develop new techniques and strategies. When machine learning models are trained on old datasets, they may not reflect current network conditions. This reduces the practical reliability of these models in today's cybersecurity settings. Models trained on outdated data may fail to capture modern encrypted traffic behavior, recent attack patterns and new anomalies linked to zero-day threats.

A further weakness in prior work is the strong reliance on supervised learning techniques. Supervised models need labeled training data, where traffic records are already categorized as normal or malicious. This is a big challenge for zero-day attack prediction because zero-day incidents are, by nature, previously unknown and usually lack labeled examples beforehand. For this reason, supervised learning alone is not ideal for detecting zero-day threats. A better approach is to use unsupervised learning, which can understand normal traffic structure without needing attack labels and can identify unusual behavior more effectively. Additionally, many earlier methods depended on single-model detection, which can limit performance. For instance, one model might recognize certain anomaly patterns well but miss others. In network traffic analysis, attack behavior can be complex and multi-dimensional, making it hard for one algorithm to catch all significant deviations. This creates a need for a stronger hybrid approach that combines the strengths of various unsupervised methods.

Another gap is that past studies often focused narrowly on one aspect of cyber threat detection, without considering it within the broader context of a real-time predictive cybersecurity system. In a complete project, cyber threat prediction is broken down into several parts, including system calls, user behavior, network traffic patterns, and endpoint logs. However, many existing studies overlook how a network-based zero-day prediction component can enhance a larger smart framework for real-time cyber threat prediction.

These limitations include the lack of direct focus on zero-day attacks, the use of outdated datasets, the insufficient use of unsupervised learning, the dependence on single-model methods, and the limited integration into broader real-time predictive system highlight the research gap that this component aims to fill.

This research addresses that gap by applying unsupervised learning models, specifically Autoencoders and Isolation Forest, to identify abnormal network traffic patterns that may suggest zero-day attacks. It also uses a modern dataset, HIKARI-2021, which is more relevant to today's network situations and attack behaviors. Furthermore, by combining these techniques in a hybrid framework, this component strives to create a more adaptable, proactive and reliable system for predicting zero-day attacks based on network traffic. This approach enhances the ability to detect subtle anomalies and supports the broader goal of building a real-time smart system for predicting cyber threats.

Research Paper	Zero-Day Prediction	Using Unsupervised Learning Techniques	Use latest and more Accurate datasets for better output
[3]		Research Gap	
[4]			Research Gap
[5]	Research Gap		
Proposed Project	Covered	Covered	Covered

1.3 Research problem

The main goal of this research study is to create a Real-Time Intelligent System for Predicting Cyber Threats. This system is divided into several parts, including predicting known attacks through system call patterns and user behavior, as well as forecasting zero-day attacks through network traffic patterns and endpoint log patterns. Among these parts, predicting zero-day attacks using network traffic patterns is particularly important. Network traffic often shows early and noticeable signs of unusual or harmful behavior.

Zero-day attacks are tough to detect because they take advantage of vulnerabilities that software developers, system admins and security vendors do not know about at the time of the attack. Since there is no patch, rule or known signature available, traditional signature-based intrusion detection systems usually cannot identify these threats in time. This poses a serious issue for modern cybersecurity. Organizations may only find out about a zero-day attack after it has caused significant damage, such as unauthorized access, service interruptions, financial loss or exposure of sensitive information. Network traffic patterns hold valuable information on how users, devices and systems communicate within a network. These patterns include packet flow behavior, connection frequency, protocol usage, communication timing, and other traffic traits that can reveal both normal and harmful activities. Legitimate users and malicious actors often behave differently at the network level, even if the attack is not yet known. For this reason, analyzing network traffic offers a strong chance to identify abnormal behavior that may be linked to zero-day threats.

However, detecting zero-day attacks through network traffic remains a challenging problem. Current systems often fall short because zero-day attacks are subtle, dynamic and can mimic normal traffic behavior. Additionally, modern network environments are complex, with large data volumes, diverse communication patterns, and constantly changing traffic conditions. This complexity makes it hard to create models that can accurately tell the difference between normal anomalies and real malicious activity. Another significant issue is that zero-day attacks typically do not have labeled examples available for training, which weakens the effectiveness of supervised learning approaches.

Therefore, the central research problem in this component is the absence of an effective and adaptable method for identifying zero-day attacks from network traffic in real time. Existing approaches often rely too much on known signatures, outdated datasets or labeled attack data, making them less suitable for detecting new and previously unseen threats. A more proactive solution is needed that can learn normal traffic behavior, recognize unusual deviations, and support early threat prediction before serious damage occurs. To tackle this problem, this research aims to apply machine learning-based anomaly detection techniques to network traffic analysis. Specifically, it focuses on using unsupervised learning methods, which are better suited for zero-day prediction since they do not require labeled attack samples. By analyzing both normal and abnormal traffic behavior, this research seeks to create a system that can identify suspicious traffic

patterns early and improve proactive cyber defense. In this way, the study contributes to the broader goal of developing a more intelligent, adaptable and real-time system for predicting cyber threats. [6]

2. Objectives

The goal of this component is to build a smart network-based system that can spot suspicious traffic linked to zero-day attacks. Since zero-day attacks don't have known signatures and there are few labeled examples, this component focuses on finding network anomalies instead of matching set attack patterns. In the overall project, this part supports the larger aim of creating a Real-Time Intelligent System for Predicting Cyber Threats, where network traffic serves as a key data source for identifying new threats. The implementation in this component uses a hybrid anomaly detection method. The system first examines network traffic data to learn normal behavior. It then generates anomaly indicators from multiple unsupervised models and combines those outputs to strengthen the final prediction. Specifically, the framework uses an Autoencoder to detect anomalies through reconstruction error, an Isolation Forest to find unusual samples through isolation-based scoring and a Logistic Regression meta-classifier to merge both outputs into a final decision. This shows that the component's goals are not just theoretical they are linked to the real implementation workflow developed for predicting zero-day attacks using network anomalies.

2.1 Main Objectives

The main aim of this research component is to create and implement a smart anomaly detection framework that can foresee potential zero-day attacks by analyzing network traffic patterns in real time or close to real time, without relying on predefined signatures or fully labeled attack data sets. In more detail, this component seeks to build a system that understands the traits of normal network behavior and identifies abnormal deviations that may signal unknown malicious actions. Since zero-day attacks are new and often lack clear labels, the system must use unsupervised learning methods to detect anomalies directly from traffic behavior. This is done by using the Autoencoder to model normal traffic patterns and the Isolation Forest to find isolated abnormal samples within the traffic data. Their outputs are then combined through a Logistic Regression-based hybrid decision layer, which makes the prediction framework stronger than depending on a single model.

Another key aspect of the main objective is to ensure that this component supports the broader project goal of developing a Real-Time Intelligent System for Predicting Cyber Threats. In this larger system, the network anomaly detection component serves as a key predictive module that helps identify unknown threats early, before serious damage can occur. Therefore, the main objective is not just to categorize traffic as normal or abnormal, but to contribute to a proactive cybersecurity approach that enhances threat prediction using network evidence.

In summary, the primary goal of this component is to develop a hybrid unsupervised machine learning framework for predicting zero-day attacks through network anomalies. This will enable the system to detect subtle, previously unseen attack behaviors more effectively than traditional signature-based methods.

2.2 Specific Objectives

To achieve the main goal, this component is divided into several specific objectives, each related to the steps used in the developed anomaly detection system.

1. To study the limitations of traditional signature-based systems for zero-day attack detection

The first objective is to examine why conventional intrusion detection methods are not enough for zero-day prediction. Traditional signature-based systems rely on known attack patterns, which means they cannot identify new threats without a created signature. This is important because it shows the need for anomaly-based and unsupervised approaches. This understanding explains why the implementation does not rely on rule matching; instead, it focuses on learning normal traffic behavior and detecting deviations.

2. To analyze network traffic patterns as a source for predicting zero-day attacks

The second objective is to use network traffic patterns to predict zero-day attacks. This involves treating traffic behavior as a valuable source of evidence about how devices and users communicate, how flows behave and how abnormal behavior can emerge during a cyberattack. The implementation is based on the idea that suspicious traffic may arise from unusual communication patterns, unexpected flow behavior or anomalies in traffic features. Therefore, the system is designed around traffic-based analyses instead of relying solely on endpoint or signature detection.

3. To prepare and preprocess network traffic data for anomaly detection

Another specific objective is to prepare the network traffic dataset for machine learning-based anomaly detection. This includes selecting a suitable dataset, removing irrelevant attributes, organizing the traffic features, and preparing the data so that the models can learn meaningful patterns. Based on the information you provided, this component uses the HIKARI-2021 dataset because it features realistic traffic behavior and is more appropriate for modern network intrusion analysis than outdated datasets. This is important since the quality and relevance of the dataset directly affects how well the models can recognize normal versus abnormal traffic.

4. To implement an Autoencoder model for learning normal network traffic behavior

A key objective of the implementation is to build an Autoencoder-based anomaly detector that learns how normal traffic should appear. In your implementation, the Autoencoder is structured as

a fully connected neural network with this architecture: input layer, dense layer with 64 neurons, dense layer with 32 neurons, bottleneck layer with 16 neurons, dense layer with 32 neurons, dense layer with 64 neurons and output layer. The model is designed to reconstruct normal traffic examples and its anomaly signal comes from reconstruction errors using Mean Squared Error. When the reconstruction error is high, the traffic is considered suspicious. This objective is important because it allows the system to identify complex abnormal patterns that do not match learned normal behavior

5. To implement an Isolation Forest model for identifying anomalous traffic records

Another specific goal is to implement Isolation Forest as a second unsupervised anomaly detector. In this component, Isolation Forest is used because it can efficiently isolate unusual data points by randomly partitioning the data through multiple trees. The implementation you described uses key hyperparameters including 300 trees, contamination = 0.01 and random state = 42. This objective supports the system by providing an alternative anomaly signal that differs from reconstruction-based detection. While the Autoencoder focuses on how well traffic can be reconstructed, Isolation Forest focuses on whether traffic points can be quickly identified as rare or abnormal

6. To combine the outputs of Autoencoder and Isolation Forest into a hybrid anomaly prediction framework

One of the main objectives is to create a hybrid model instead of relying on a single unsupervised detector. In your system, the Autoencoder produces AE_error, while Isolation Forest generates IF_score. These two outputs are combined into a joint feature vector

Hybrid_Input = [AE_error, IF_score]

This objective is significant because it reflects the actual system design you implemented. The goal is to leverage the strengths of both models together. The Autoencoder excels at capturing complex traffic patterns, while Isolation Forest is efficient at isolating outlier behavior. By combining them, the system becomes more reliable for identifying subtle zero-day anomalies.

7. To train a Logistic Regression meta-classifier for final anomaly decision making

Another specific objective is to create a meta-classification stage that learns how to combine the anomaly outputs of the two base models. In your implementation, this uses Logistic Regression. The meta-classifier is trained on the combined feature vector built from Autoencoder reconstruction error and Isolation Forest score. It then calculates the probability that a traffic sample is anomalous using predict_proba(X_meta). Next, the probability is compared against a predefined threshold to classify the sample as either Anomaly or Normal. This objective is crucial because it transforms two separate anomaly indicators into a single final prediction mechanism and improves the quality of the overall detection process

8. To develop a prediction mechanism that works without depending on labeled zero-day attack samples

The core objective of this research is to create a system that can function even when labeled zero-day attack data is unavailable. This is one of the biggest challenges in zero-day prediction. Since new attacks are unknown in advance, labeled examples are rare or nonexistent. Therefore, the implementation relies mainly on unsupervised learning, which does not require complete attack labels to find anomalies. This objective directly explains why your component uses Autoencoder and Isolation Forest instead of only supervised classification methods.

9. To improve the adaptability of the cyber threat prediction system by using a modern and relevant dataset

Another specific objective is to ensure that the developed model is trained and tested on a dataset that better reflects modern network environments. Your component addresses the shortcomings of earlier studies that used outdated datasets by selecting HIKARI-2021. This is important because modern traffic contains more realistic communication behavior, diverse attack situations and newer patterns that older benchmark datasets may not represent. Using a recent dataset enhances the relevance of the anomaly detection framework and makes the model more suitable for today's cybersecurity challenges.

10. To contribute a network-based zero-day prediction module to the overall Real-Time Intelligent System for Predicting Cyber Threats

The final specific objective is to make sure this component functions as a meaningful submodule within the larger project. The full research system includes multiple subcomponents such as system call analysis, user behavior analysis, network traffic analysis, and endpoint log analysis. This network anomaly detection piece specifically contributes to predicting zero-day attacks through traffic behavior, making it an essential part of the overall real-time cyber threat intelligence framework. This objective highlights that the implementation is not standalone; it is intended to enhance the overall research system by providing specialized and proactive traffic-based prediction capabilities.

3. Methodology

Zero-day attacks are often unknown and typically lack reliable examples. Therefore, the methodology focused on unsupervised anomaly detection instead of traditional supervised classification. The overall implementation followed a clear workflow that included data collection, traffic data preparation, preprocessing, developing individual anomaly detection models, combining model outputs into a hybrid prediction framework, and evaluating performance. This part was implemented as part of the broader Real-Time Intelligent System for Predicting Cyber Threats, but its specific task was to analyze network traffic behavior and detect suspicious deviations that may point to zero-day threats. The implementation used two main unsupervised methods: Autoencoder and Isolation Forest; it then combined their outputs with a Logistic Regression meta-classifier. This design choice was made because a single anomaly detector might not capture all types of network anomalies, especially in complex and changing traffic environments. By using two different detection methods, the goal was to create a more reliable and adaptable system for predicting zero-day attacks.

3.1 Data Collection and Preprocessing

The first step of the methodology involved finding and collecting a suitable dataset for network anomaly detection. Since this component aims to predict zero-day attacks through traffic behavior, the chosen dataset needed to represent realistic network traffic patterns, include a good mix of normal and malicious behaviors, and have enough variety to support anomaly-based learning. Based on these requirements, the implementation used the HIKARI-2021 dataset. According to the information you provided, this dataset is particularly suitable because it combines realistic network traffic with malicious scenarios, including encrypted communications, and it reflects modern conditions for intrusion detection better than older datasets. Choosing HIKARI-2021 was a key decision because many previous studies relied on outdated datasets that do not accurately represent current network behavior. In contrast, HIKARI-2021 offers a richer and more modern traffic environment, making it more fitting for training and testing unsupervised models designed to identify subtle anomalies linked to zero-day attacks. This relevance makes the data collection stage crucial for improving real-world applicability. After collecting, the network traffic data was prepared for the anomaly detection process. This preparation involved organizing the traffic records into a format that machine learning algorithms could handle. Since network traffic datasets may include a mix of numerical attributes, categorical information, identifiers, and other metadata, the raw data needed careful preparation. The main goal at this stage was to ensure that the data accurately reflected traffic behavior in a clean, structured, and usable format. The dataset was also selected to support both models in the hybrid framework. For the Autoencoder, it provided enough examples of normal traffic behavior to help the model learn the expected structure of legitimate communication. For Isolation Forest, the diversity and variability in the traffic helped the model identify rare or unusual patterns more effectively. This means the data collection phase aimed not just at obtaining any dataset but at selecting one that supports the specific strategy of combining reconstruction-based and isolation-based anomaly detection. Another important aspect was

ensuring the collected data fit the broader purpose of the project. Since this component is part of a larger intelligent threat prediction system, the network traffic data needed to realistically detect suspicious cyber behavior in a way that could integrate with other predictive modules such as user behavior, system calls, and endpoint logs. Thus, the data collection step was designed not just for isolated model training, but for constructing a network-based prediction component that fits into the larger cybersecurity framework.

3.2 Data Preprocessing

Data preprocessing was one of the most critical parts of the implementation because raw network traffic is usually not immediately suitable for anomaly detection models. Preprocessing was essential to convert the collected traffic data into a format the Autoencoder and Isolation Forest algorithms could be used effectively. The quality of the preprocessing stage directly affects how well the final system can tell the difference between normal traffic behavior and suspicious anomalies. The first preprocessing task involved cleaning the dataset. This included checking for missing values, duplicate records, inconsistent feature representations, and noisy data that might negatively affect model learning. If such problems remain in the dataset, anomaly detection models could treat data quality issues as real anomalies, which would lower reliability. Therefore, the preprocessing workflow aimed to reduce noise and improve consistency before any model training began. The next step involved selecting relevant traffic features. In network traffic data, not all attributes contribute equally to anomaly detection. Some features may be identifiers, administrative fields, or attributes that don't meaningfully describe behavior. Retaining such fields unnecessarily can reduce model generalization. Therefore, the preprocessing stage focused on keeping the traffic features most useful for representing communication behavior, flow patterns, and suspicious deviations. This is especially important for unsupervised learning, where models depend heavily on the structure and quality of the input features rather than on external labels. Another major preprocessing activity was transforming the data into a form suitable for the models. Machine learning models like Autoencoders and Isolation Forest require structured numeric input. Consequently, categorical traffic-related attributes, if present, must be converted into numerical form. This ensures that the models can process all features consistently and learn meaningful relationships across different traffic dimensions. The methodology also needed to scale or normalize feature values. Network traffic datasets often contain attributes with very different ranges. Some features may have very large numerical values, while others may be small. Directly using these varying scales can cause larger features to dominate the learning process, particularly in neural network-based models like Autoencoders. Normalization helps ensure that the models treat all relevant features more fairly and stabilizes the training process. Since this component focuses on detecting anomalies, preprocessing also supports distinguishing between normal traffic learning and anomaly evaluation. In the Autoencoder stage, the goal is to learn the structure of normal traffic so that reconstruction errors can later reveal suspicious deviations. In the hybrid model, the anomaly generated indicators from both the Autoencoder and Isolation Forest are combined for final prediction. Therefore, preprocessing was designed to maintain behavioral

consistency in the input space, allowing both models to produce useful anomaly signals. Another important point is that preprocessing was not performed merely for general classification; it was specifically designed for zero-day prediction through anomaly analysis. This means the preprocessing pipeline had to preserve subtle differences in network behavior instead of oversimplifying the data. Since zero-day attacks may manifest as small deviations from normal traffic, the feature preparation process needed to retain enough behavioral detail for the unsupervised models to identify those deviations. Overall, the preprocessing stage transformed HIKARI-2021 traffic data into a structured, clean, normalized and model-compatible dataset that could support both unsupervised detectors and the later hybrid decision stage. This step laid the groundwork for the rest of the implementation.

3.3 Model Architecture

The model architecture uses a combined anomaly detection approach made up of three linked components: an Autoencoder, an Isolation Forest, and a Logistic Regression meta-classifier. This structure was designed to improve zero-day attack prediction by merging the benefits of deep learning for traffic reconstruction and ensemble methods for anomaly detection. You can find a detailed explanation of the workflow in the PDF you provided.

Autoencoder Architecture

The first part of the architecture is the Autoencoder model. This Autoencoder is a fully connected neural network with an encoder and a decoder. The encoder compresses the original traffic features into a lower-dimensional hidden representation. The decoder then reconstructs the original input from this compressed form. As outlined in your PDF, the architecture includes:

- Input Layer
- Dense Layer with 64 neurons
- Dense Layer with 32 neurons
- Bottleneck Layer with 16 neurons
- Dense Layer with 32 neurons
- Dense Layer with 64 neurons
- Output Layer

This design was chosen to lower the dimensionality of the input and push the network to learn the essential patterns of normal network traffic. The bottleneck layer serves as a compact representation of the original traffic. During training, the Autoencoder learns to recreate normal traffic samples with minimal errors. When it later receives unusual traffic that strays from the learned normal pattern, it generates a higher reconstruction error. [7]

```
def build_autoencoder(input_dim: int):
    inp = keras.Input(shape=(input_dim,))
    x = layers.Dense(64, activation="relu")(inp)
    x = layers.Dense(32, activation="relu")(x)
    bottleneck = layers.Dense(16, activation="relu")(x)
    x = layers.Dense(32, activation="relu")(bottleneck)
    x = layers.Dense(64, activation="relu")(x)
    out = layers.Dense(input_dim, activation="linear")(x)
    model = keras.Model(inp, out)
    return model
```

The anomaly signal from the Autoencoder relies on the Mean Squared Error (MSE) between the original input and the reconstructed output

$$\text{Error} = \text{mean}((X - X_{\text{reconstructed}})^2)$$

If the reconstruction error goes beyond a set threshold, the traffic sample is marked as anomalous. This feature makes the Autoencoder useful for zero-day prediction because it does not need exact labels for unknown attacks. Instead, it relies on the model's ability to learn normal traffic and highlight deviations.

Isolation Forest Architecture

The second part of the architecture is the Isolation Forest model. The Isolation Forest is an ensemble-based unsupervised anomaly detection algorithm that finds anomalies by isolating samples through random partitioning. The method is based on the idea that unusual data points are easier to separate from normal points. Generally, anomalies have shorter path lengths within the random tree structure. [8]

In your implementation, the Isolation Forest model uses the following main hyperparameters:

- Number of Trees: 300
- Contamination: 0.01
- Random State: 42

```
# Good starting params for anomaly detection
N_ESTIMATORS = 300
CONTAMINATION = 0.01
RANDOM_STATE = 42
N_JOBS = -1

def main():
    os.makedirs(MODEL_DIR, exist_ok=True)
```

These parameters help manage how the model detects anomalies and estimates the proportion of abnormal observations in the data. The model generates an anomaly score through its decision function, where lower scores signal more suspicious traffic behavior. The Isolation Forest is valuable in this part because it efficiently handles high-dimensional traffic data and can identify rare or unusual observations without fully knowing the attack patterns.

Hybrid Integration Architecture

The key aspect of the model design is the hybrid integration stage. This stage was included because the Autoencoder and Isolation Forest capture different parts of abnormal traffic behavior. The Autoencoder excels at identifying deviations from learned normal patterns, while the Isolation Forest is stronger at isolating statistically unusual points. Since neither model is perfect alone, their outputs are merged to enhance the final prediction process.

The hybrid process involves these steps:

1. The Autoencoder produces a reconstruction error value for each traffic sample.
2. The Isolation Forest generates an anomaly score for each sample via its decision function.
3. The outputs are combined into a feature vector:

Hybrid_Input = [AE_error, IF_score]

4. This combined feature vector is then passed to a Logistic Regression meta-classifier.
5. The meta-classifier learns how to merge these two anomaly indicators to create the final output.

Meta-Classifer Decision Layer

The final layer of the architecture is the Logistic Regression meta-classifier, which serves as the decision-making point of the hybrid system. Its role is to take the outputs from the Autoencoder and Isolation Forest and calculate the probability that a given traffic sample is anomalous. According to the PDF, this is accomplished using the function:

```
meta_model.predict_proba(X_meta)
```

The classifier then compares the anomaly probability to a set decision threshold. If the probability is equal to or above that threshold, the sample is considered an Anomaly. Otherwise, it's classified as Normal. This architectural design is a significant strength of the implementation. It combines two distinct anomaly detection outputs into a single, unified decision. This flexibility enhances the system's ability to detect complex zero-day attack behavior hidden in network traffic.

```

# Load base models
ae = tf.keras.models.load_model(os.path.join(AE_DIR, "model.h5"))
with open(os.path.join(AE_DIR, "threshold.json"), "r", encoding="utf-8") as f:
    ae_thr = float(json.load(f)["threshold_value"])

iso = joblib.load(os.path.join(IF_DIR, "model.pkl"))

# Base model scores
ae_err = ae_recon_error(ae, X_val)
if_score_normality = iso.decision_function(X_val)
if_anom = -if_score_normality

# Optional binary flags
ae_flag = (ae_err > ae_thr).astype(int)
if_flag = (iso.predict(X_val) == -1).astype(int)

# Hybrid features (stacking)
X_meta = np.column_stack([ae_err, if_anom, ae_flag, if_flag]).astype(np.float32)

```

Technologies Used

The implementation of the component, Predicting Zero-Day Attacks Using Network Traffic Patterns, involved several technologies, programming libraries and machine learning frameworks to develop, train, and evaluate the anomaly detection models. These technologies assisted with data preprocessing, model building, anomaly detection, hybrid model integration, and evaluation. Below are the main technologies used.

Python

Python served as the primary programming language for the entire implementation. It was used for handling datasets, preprocessing, model training, testing, and evaluation. Python was chosen because it offers strong support for data science, machine learning and cybersecurity research.

TensorFlow / Keras

TensorFlow with Keras was utilized to build and train the Autoencoder model. Keras was particularly helpful because it provides a straightforward way to design deep learning structures. In this component, it defined the encoder and decoder layers of the Autoencoder, trained the network on normal traffic patterns, and calculated reconstruction errors for anomaly detection.

Scikit-learn

Scikit-learn played a crucial role in several important parts of the implementation. It was used to:

- implement the Isolation Forest anomaly detection model,

- implement the Logistic Regression meta-classifier,
- support preprocessing steps like scaling,
- calculating evaluation metrics such as accuracy, precision, recall and F1-score.

This library was essential because it provided efficient and reliable machine learning tools for the hybrid model pipeline.

Pandas

Pandas were used for loading, organizing, cleaning, and manipulating the HIKARI-2021 dataset. It assisted in reading the dataset, selecting relevant columns, handling missing values, and preparing the traffic data before training the models.

NumPy

NumPy supported numerical computations throughout the implementation. It facilitated matrix operations, array handling, calculation of reconstruction error values, and preparation of model input and output data.

3.4 Evaluation

The evaluation stage measured how effectively the implemented framework could differentiate between normal traffic and suspicious anomalies. Given that this component focuses on predicting zero-day attacks via network anomalies, evaluation was vital in assessing whether the developed models were generating practical anomaly decisions.

The framework was evaluated using standard performance metrics common in cyber threat detection and machine learning. According to the PDF, these metrics include:

- Accuracy
- Precision
- Recall
- F1-score

These metrics are essential because the performance of anomaly detection cannot be assessed using a single value. Accuracy offers a general sense of how often the system makes correct predictions, but it may not adequately reflect performance when normal and anomalous classes are imbalanced. Precision indicates how many predicted anomalies are genuinely suspicious, addressing the false positive issue. Recall is crucial for showing how many real anomalous cases the system detects. The F1-score balances precision and recall, making it particularly useful when evaluating threat detection systems where missed attacks and false alarms are significant concerns.

The methodology evaluated the individual Autoencoder model, the individual Isolation Forest model, and the final hybrid model. This separate assessment was critical as it allowed for comparing the impact of each model and determining if hybrid integration genuinely enhanced the

anomaly detection process. The evaluation stage not only assessed final system performance but validated the need for a combined architecture instead of relying on a single unsupervised method.

Another important aspect of the evaluation in this methodology is the role of the decision threshold in the meta-classifier stage. Since the final prediction depends on the anomaly probability produced by Logistic Regression, the threshold influences how strictly the system classifies traffic. This is significant because choosing the threshold impacts the balance between detecting more suspicious activity and minimizing unnecessary false positives. In real cyber threat detection, this balance is crucial because overly sensitive detection may flood analysts with alerts, while being too strict may overlook subtle zero-day patterns. [9]

Overall, the evaluation methodology aimed to test if the hybrid anomaly detection framework could reliably predict zero-day attacks from network traffic patterns. By measuring the base models' behavior and the final hybrid decision layer using various metrics, this stage helped confirm whether the implementation is appropriate for use as a proactive module in a broader real-time cyber threat prediction system.

4. Results and Discussions

This section describes how the anomaly detection framework predicted zero-day attacks by analyzing network traffic patterns. We evaluated the implementation with three model configurations: the Autoencoder model, the Isolation Forest model and the Hybrid model, which combined the outputs of the Autoencoder and Isolation Forest using a Logistic Regression meta-classifier. This evaluation aimed to see how well the framework could spot suspicious traffic behavior that might signal previously unknown cyber threats.

The results indicate that each model had useful anomaly detection capabilities, while the hybrid method showed the best overall performance. This aligns with the goal of creating a more flexible and proactive way to detect zero-day attacks through network anomaly analysis. The figures produced during the implementation further illustrate the training behavior, anomaly separation, classification performance, and metric-based comparison of the models.

4.1 Results

The Autoencoder model achieved an accuracy of 85.27%, precision of 29.95%, recall of 87.35% and F1-score of 44.60%. These results show that the Autoencoder effectively learned normal traffic behavior and identified unusual deviations through reconstruction error. The high recall value indicates that the model was very responsive in spotting suspicious traffic instances, which is a key quality in predicting zero-day attacks.

Figure 4.1 illustrates the Autoencoder training process.

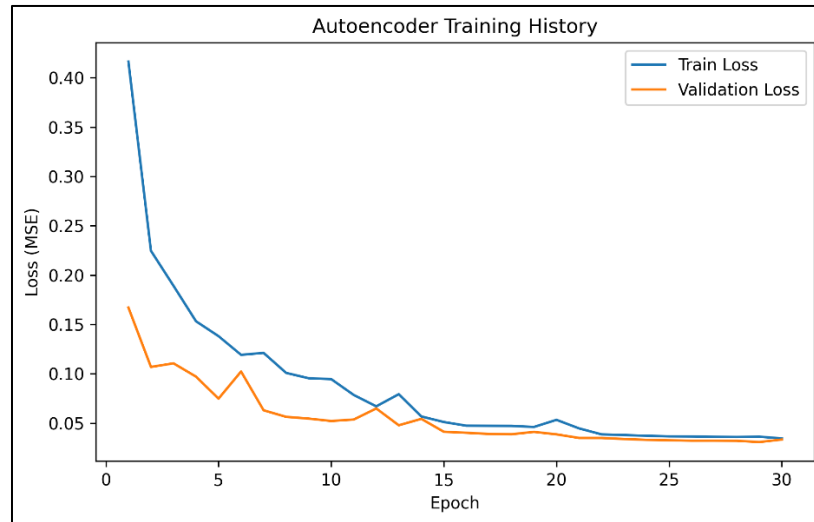


Figure 4.1

This figure should follow the Autoencoder results paragraph because it visually shows how the model learned during training and validation.

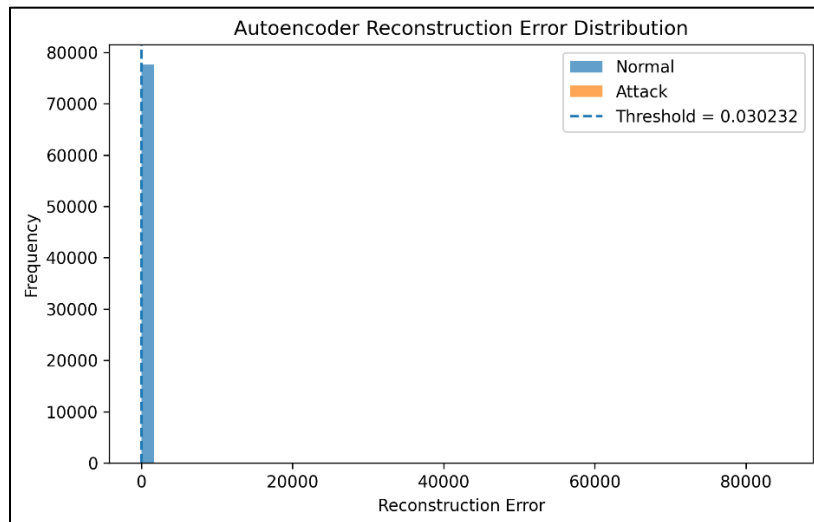


Figure 4.2

Next, we can use Figure 4.2, to display the model's anomaly separation behavior, which is appropriate to place after the training history figure. This figure supports the discussion by showing how reconstruction errors were distributed and how we could distinguish anomalous traffic through the selected threshold.

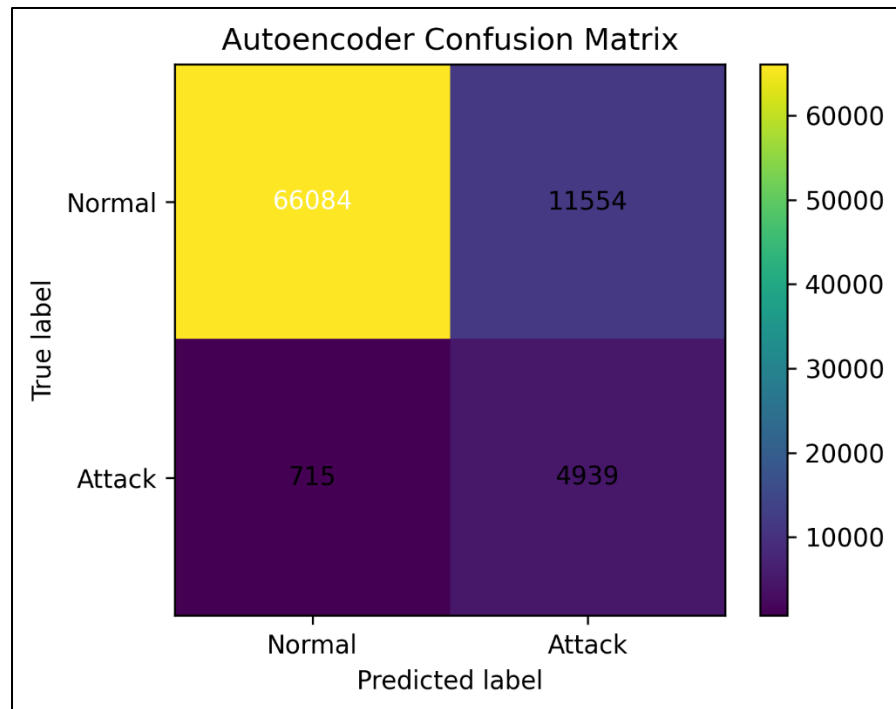


Figure 4.3

Figure 4.3 can explain the Autoencoder's classification behavior.

This figure fits best after discussing anomaly separation because it clearly shows how the model classified normal and anomalous samples. Additionally, we can further present the predictive quality of the Autoencoder using Figure 4.4 and Figure 4.5. These figures are best at the end of the Autoencoder subsection because they provide formal graphical support for the model's classification ability.

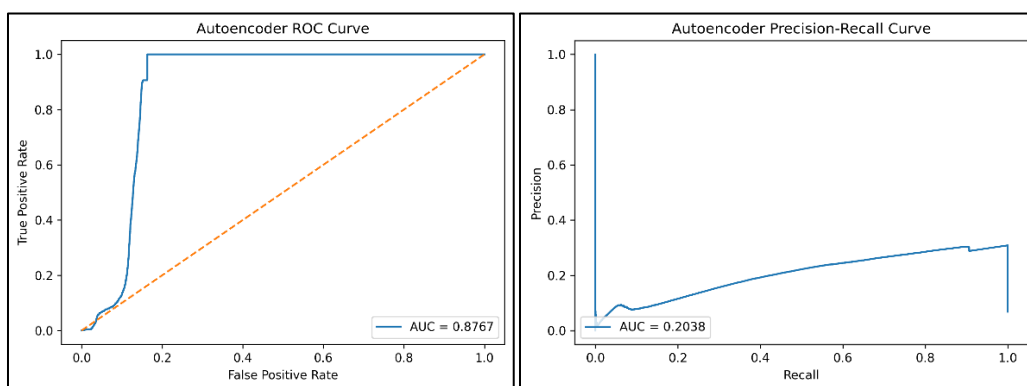


Figure 4.4 and Figure 4.5

The Isolation Forest model had an accuracy of 84.97%, precision of 9.14%, recall of 13.58%, and F1-score of 10.93%. These results demonstrate that the model could identify unusual traffic points using isolation-based anomaly detection. As an unsupervised model, it offered an important

alternative perspective by focusing on rare and distinct traffic patterns instead of reconstruction behavior. This makes the Isolation Forest a vital component of the hybrid framework, complementing the deep learning-based detection of the Autoencoder.

We can present the prediction behavior of the Isolation Forest model using Figure 4.6.

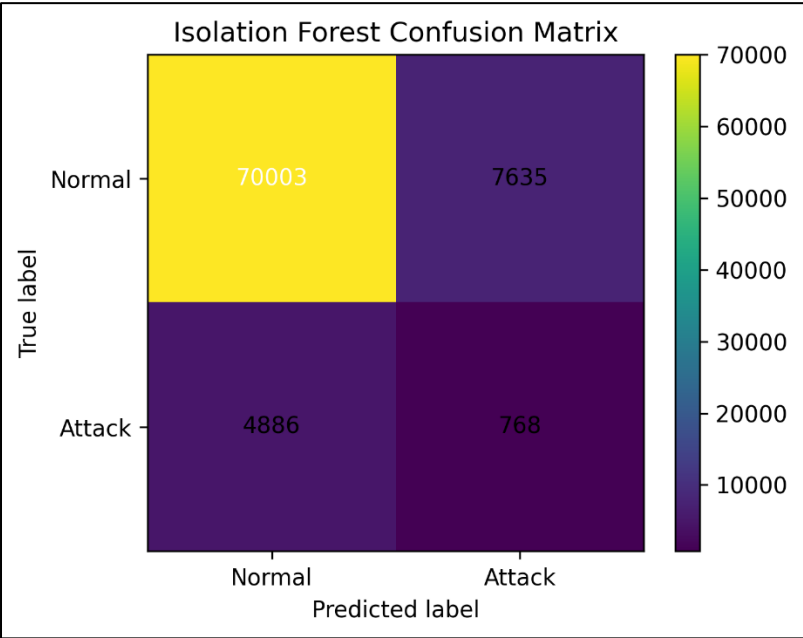


Figure 4.6

This figure should immediately follow the Isolation Forest results paragraph because it formally shows the model's classification outcomes. To provide a comprehensive visual presentation of the model's anomaly detection features, we can include Figure 4.7 and Figure 4.8. These figures fit best after the confusion matrix discussion.

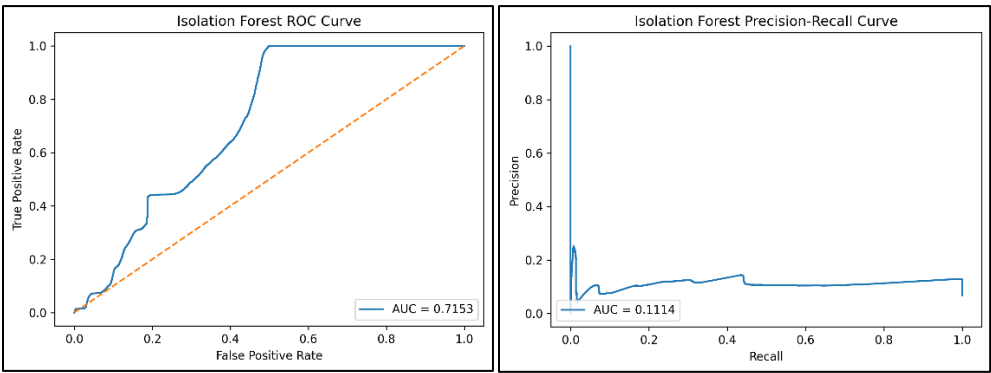


Figure 4.7 and Figure 4.8

The Hybrid model outperformed the others with an accuracy of 92.12%, precision of 45.15%, recall of 75.01% and F1-score of 56.37%. These results confirm that combining the outputs of the Autoencoder and Isolation Forest through a Logistic Regression meta-classifier led to a more

robust prediction framework. The hybrid model provided the highest accuracy and F1-score among the tested approaches, showing that the combined framework was more effective in recognizing network anomalies related to zero-day threats. This outcome strongly supports the choice of using hybrid unsupervised architecture instead of relying on a single anomaly detector.

We can illustrate the performance of the hybrid classifier with Figure 4.9.

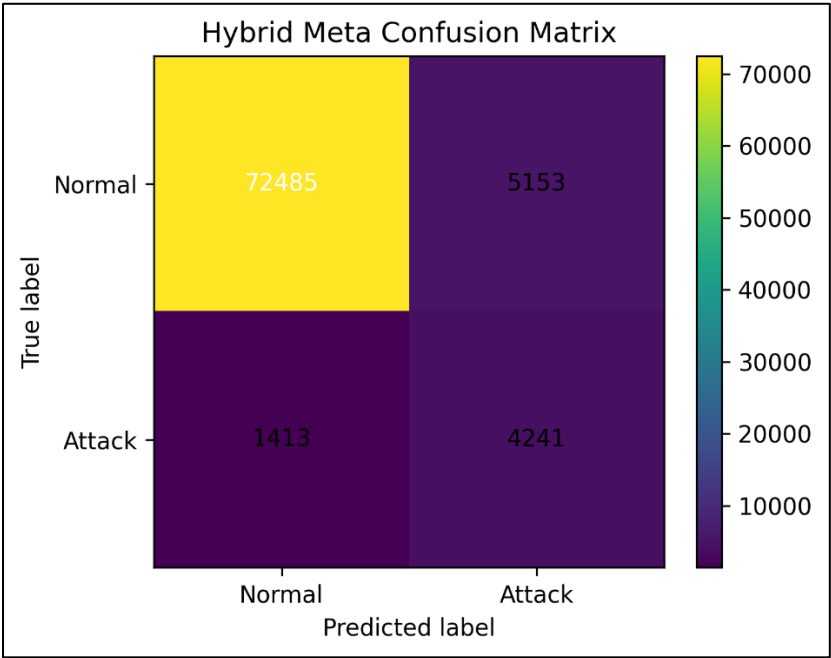


Figure 4.9

This figure is most suitable right after the hybrid model results paragraph because it visually supports the classification quality of the final system. We can then include additional performance analysis using Figure 4.10 and Figure 4.11. These figures should be placed immediately after the confusion matrix figure for a complete graphical view of the final model’s predictive strength.

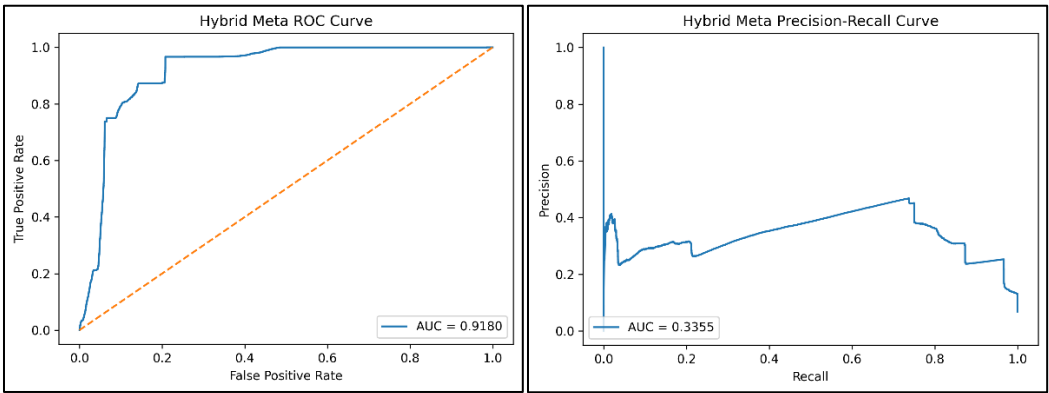


Figure 4.10 and Figure 4.11

Finally, we can present a direct comparison of all three models using metric comparison charts generated during the implementation.

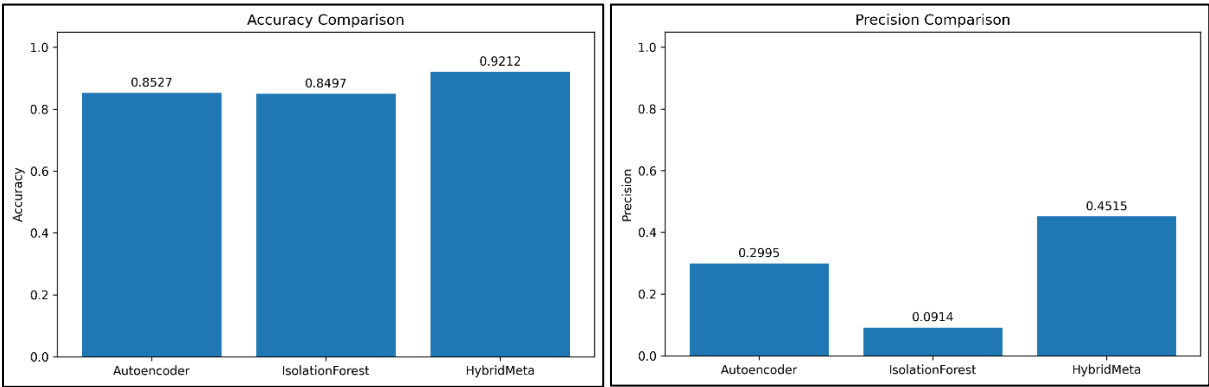


Figure 4.12 and Figure 4.13

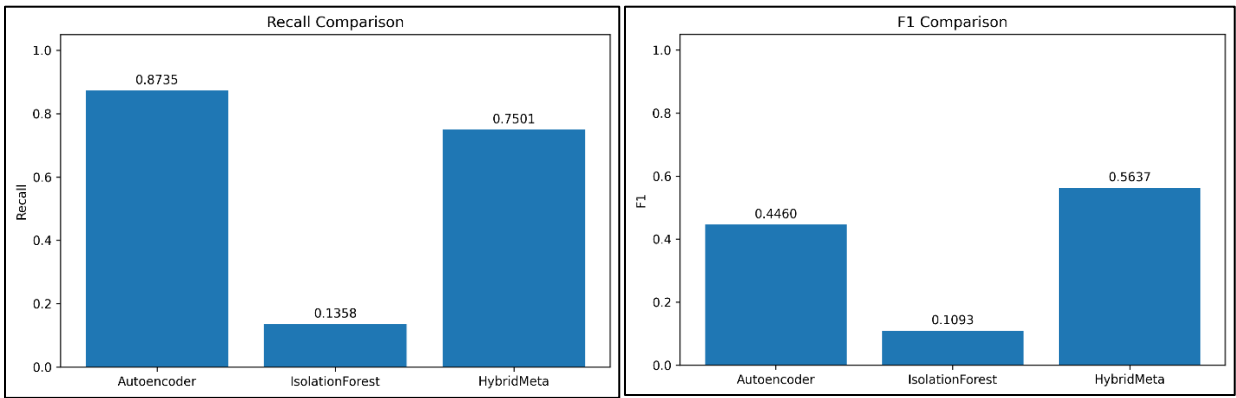


Figure 4.14 and Figure 4.15

These summarize the comparative performance of the Autoencoder, Isolation Forest and Hybrid models clearly and formally. Together, these comparison figures show that the hybrid approach provided the strongest overall performance across the evaluated metrics.

4.2 Research Findings

The findings of this research confirm that network traffic patterns are a strong basis for predicting zero-day attacks through anomaly detection. By analyzing traffic behavior instead of relying on predefined signatures, the framework was able to identify suspicious activity linked to previously unseen cyber threats. This is crucial for predicting zero-day attacks, as such threats often aren't represented by existing rules or known signatures. A key finding of the study is that unsupervised learning is well-suited for this problem. Zero-day attacks generally lack labeled training samples, so unsupervised models offer a practical solution by learning normal behavior and highlighting deviations. In this case, the Autoencoder enabled deep reconstruction-based anomaly learning, while Isolation Forest provided isolation-based anomaly scoring. Their integration through the

hybrid model demonstrated the value of combining multiple unsupervised anomaly signals into a unified decision framework.

Another important finding is that a hybrid anomaly detection framework is more effective than relying on a single method. The experimental performance values showed that integrating the outputs of the Autoencoder and Isolation Forest through Logistic Regression resulted in a more precise and trustworthy final prediction mechanism. This suggests that reconstruction-based and isolation-based anomaly indicators complement each other well in the realm of network traffic analysis. The study also highlights the importance of using a modern and relevant dataset for implementation. The use of HIKARI-2021 supported the development of a model that aligns better with contemporary network environments and realistic traffic behavior. This is significant for improving the practical relevance of zero-day prediction research and for avoiding the drawbacks linked to outdated traffic datasets. [10]

Overall, the research findings show that the implemented hybrid framework is a meaningful contribution to the broader Real-Time Intelligent System for Predicting Cyber Threats, especially within the subcomponent focused on predicting zero-day attacks using network traffic patterns. The implementation indicates that anomaly-based traffic analysis can provide a solid foundation for proactive cyber threat prediction.

4.3 Challenges

One major challenge in this research was the difficulty of predicting zero-day attacks since these attacks involve unknown vulnerabilities and lack predefined signatures. This complexity exceeds traditional intrusion detection and demands models that can detect subtle abnormal behavior directly from traffic patterns rather than known attack definitions. This challenge is a key reason why anomaly detection and unsupervised learning were chosen for the implementation. Another important challenge was dealing with the complexity of modern network traffic. Network environments include various communication types, diverse protocols, encrypted traffic, and constantly changing usage patterns. Therefore, the model had to learn from highly variable behavior while also recognizing meaningful anomalies. Preparing data that maintained this useful traffic behavior for anomaly analysis was a significant part of the implementation process.

A further challenge involved merging multiple anomaly indicators into one final prediction mechanism. Since the Autoencoder and Isolation Forest produce different types of anomaly-related outputs, the hybrid framework required a suitable integration strategy. This was addressed by using a Logistic Regression meta-classifier, which effectively combined both outputs and transformed them into a final anomaly probability. Designing this integration layer was a crucial part of the methodology and the final system architecture.

Another challenge was selecting a framework that would remain relevant within the broader project context. Since this component is just one part of the complete real-time cyber threat prediction system, its implementation needed to be suitable for future integration with other

subcomponents, including system calls, user behavior and endpoint logs. Ensuring that the network anomaly prediction model could work as a meaningful and scalable module within the larger system was another important practical challenge.

4.4 Future Implementations

A significant future direction is extending this framework into a real-time network monitoring environment. While the current implementation showcases the effectiveness of the hybrid model using prepared network traffic data, a future version of the system might continuously analyze live traffic streams and generate anomaly-based alerts in operational environments. This would enhance the practical value of the component within the overall real-time cyber threat prediction system. Another key future direction is including additional machine learning and deep learning models in the hybrid structure. Expanding the framework with more advanced anomaly detection methods could capture a wider range of traffic irregularities and further improve zero-day prediction performance. This could involve utilizing more complex neural models, sequence-aware approaches or extra ensemble-based anomaly detectors.

Future implementation can also focus on improving alert explanations and analyst support. In practical cybersecurity environments, it is important not only to detect anomalies but also to provide interpretable reasoning about why traffic was flagged as suspicious. Thus, future enhancements might incorporate explainability features that highlight influential traffic attributes or offer more informative summaries for human investigators. Another valuable future step is integrating this network traffic prediction component with other subcomponents of the complete project. Since the overall research aims to build a Real-Time Intelligent System for Predicting Cyber Threats using various sources such as system calls, user behaviors, network patterns and endpoint logs, combining these sources into one intelligent framework would significantly enhance overall cyber threat prediction capability. In such a system, the network anomaly detection component would serve as a vital predictive layer among several coordinated modules.

Finally, future work may involve refining the hybrid decision mechanism through improved threshold selection, adaptive learning strategies and more sophisticated score fusion methods. These advancements would support the long-term goal of creating a more intelligent, scalable and operationally effective framework for predicting zero-day attacks in modern network environments.

5. Testing and Implementation

The testing and implementation phase of this research was carried out to confirm that the proposed anomaly detection framework could effectively predict zero-day attacks using network traffic patterns. Since the main goal of this component was to create a practical and intelligent prediction mechanism rather than just a theoretical model, implementation was crucial to the study. The developed system needed to process network traffic data, prepare it for machine learning, apply unsupervised anomaly detection models, combine their outputs, and generate final predictions that could help identify suspicious cyber activity early.

This component was part of the larger Real-Time Intelligent System for Predicting Cyber Threats, where the network traffic module is a key subcomponent responsible for identifying potential zero-day threats. In this broader context, the implementation needed to be designed not only to support anomaly detection on its own but also to contribute to a larger predictive security system. This meant that the network traffic anomaly prediction component had to be structured, reliable, and suitable for integration with other modules of the overall system, including system calls, user behaviors, and endpoint log patterns. The implementation process began with the HIKARI-2021 dataset, chosen for its modern approach to network intrusion data. This dataset features realistic traffic behavior, diverse traffic elements, and a mix of normal and malicious patterns. It provided a solid foundation for building anomaly detection models because it better reflects current network environments than many older datasets used in past studies. Once the dataset was selected, the implementation focused on preparing the traffic data for anomaly detection, ensuring the input features were suitable for unsupervised learning models.

The main implementation framework consisted of three connected stages. The first stage involved the Autoencoder, implemented as a fully connected deep learning model that learned normal traffic behavior through reconstruction. The second stage used the Isolation Forest, which identified unusual traffic samples through isolation-based anomaly detection. The third stage was the Logistic Regression meta-classifier, which combined the outputs of the first two models and produced the final anomaly decision. This hybrid arrangement was necessary because relying on one method alone was not sufficient for the complexity of predicting zero-day attacks. By using both reconstruction-based and isolation-based anomaly signals, the system could create a more complete predictive framework. From an implementation standpoint, the Autoencoder was designed with an encoder-decoder structure. It contained an input layer, hidden dense layers with 64 and 32 neurons, a bottleneck layer of 16 neurons, and mirrored decoder layers with 32 and 64 neurons before the output layer. This architecture helped the model compress the normal traffic feature space into a lower-dimensional representation and then reconstruct it. The reconstruction error, measured using Mean Squared Error, served as the anomaly indicator. During testing, traffic samples with higher reconstruction errors were treated as more suspicious.

The Isolation Forest used an ensemble approach with 300 trees, a contamination level of 0.01, and random state 42, as outlined in your project document. This model provided an independent anomaly score based on its ability to isolate unusual traffic samples through random partitioning. Its role was significant since it offered a different analytical view than the Autoencoder. While the Autoencoder focused on deviations from learned normal structures, the Isolation Forest emphasized the rarity and separation of data points within the traffic space. The most critical stage of the implementation was the hybrid integration layer. In this step, the Autoencoder reconstruction error and the Isolation Forest score were combined into a feature vector and then passed to the Logistic Regression meta-classifier. This meta-classifier generated the final probability of anomaly for each traffic sample. This probability was then compared to a decision threshold, classifying each sample as either Normal or Anomaly. This implementation strategy created a decision-making system that was more structured and better suited for predicting zero-day attacks than using either model in isolation.

Testing was conducted to ensure that the implemented models performed as expected at each stage of the pipeline. The Autoencoder was tested to verify that it could learn the normal characteristics of the traffic data and generate meaningful reconstruction error values. The Isolation Forest was tested to confirm that it could produce anomaly-related scores for the same traffic samples. The hybrid stage was then tested to ensure the outputs of both models were successfully combined to generate final anomaly probabilities. These tests were essential for confirming that the full prediction pipeline functioned correctly from input to output. The implementation was also evaluated using standard performance metrics. The models were assessed based on accuracy, precision, recall, and F1-score, which are vital for anomaly detection and predicting cyber threats. These metrics provided a clear way to examine how effectively the model distinguishes suspicious traffic from normal traffic. They particularly helped confirm whether the final system could serve as a practical anomaly-based predictor in a cybersecurity context.

Another important aspect of the implementation was generating visual outputs like training history graphs, reconstruction error distributions, confusion matrices, ROC curves, Precision-Recall curves, and metric comparison charts. These outputs were valuable for model analysis and implementation testing, providing evidence that each stage of the anomaly detection framework was functioning correctly. For example, the training history confirmed the Autoencoder's learning behavior, while the confusion matrices and comparison charts illustrated the models' performance during evaluation. These visual outputs enhanced the testing process by making the implementation more observable and understandable.

Overall, the testing and implementation phase showed that the developed framework could process network traffic data, generate anomaly indicators, integrate multiple unsupervised outputs, and produce final anomaly decisions through the hybrid model. This phase demonstrated that the component was not only conceptually valid but also practically implemented in a way that supports the main research goal of predicting zero-day attacks through network traffic anomalies.

5.1 Overview

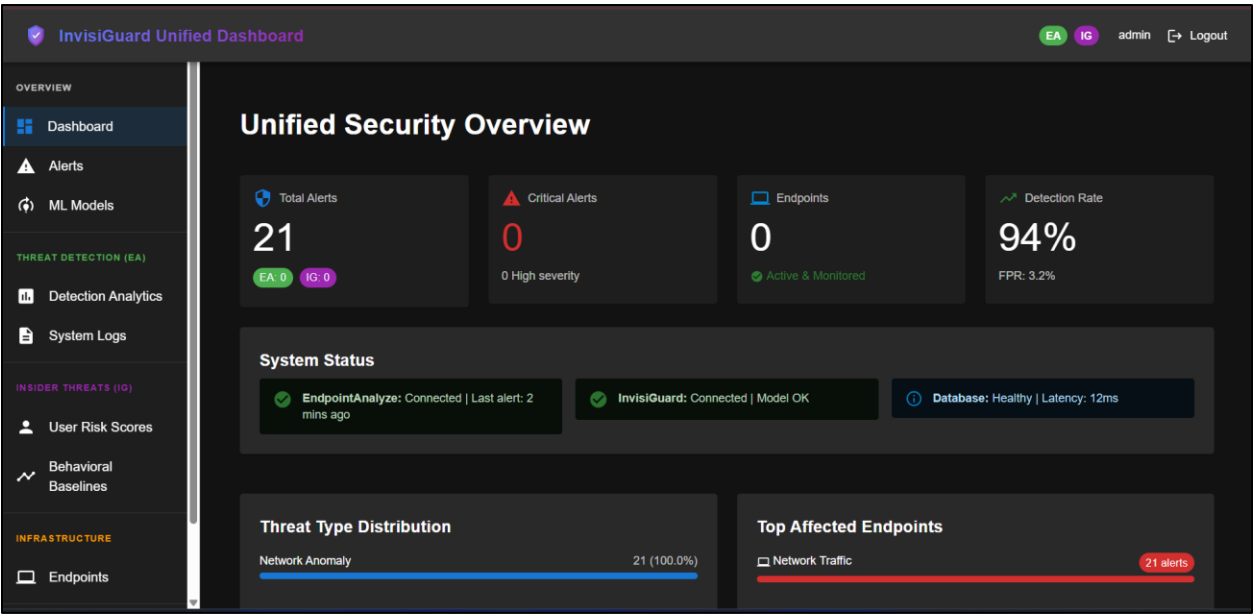
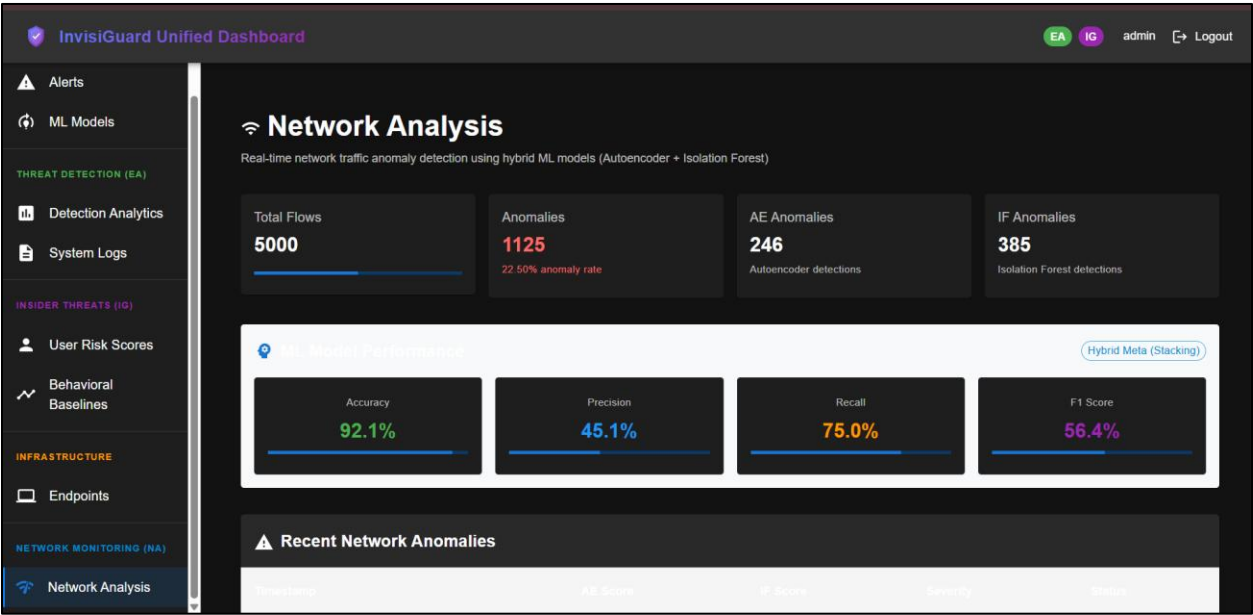
The testing and implementation process consists of connected stages that start with data preparation and end with predicting final anomalies. First, the network traffic data from HIKARI-2021 was prepared and changed into a suitable format for detecting anomalies. This ensured that the data used accounted for meaningful traffic behavior and was suitable for machine learning analysis.

Next, the Autoencoder model was developed and trained. This stage aimed to learn normal traffic patterns so that later, abnormal traffic could be recognized by a high reconstruction error. Training and validating the Autoencoder was a significant early step because it formed the basis for reconstruction-based anomaly detection. After training, the model tested traffic data and calculated reconstruction errors for each sample. These errors served as the first indication of anomalies in the framework. After that, the Isolation Forest model was used as the second detector of anomalies. This stage aimed to find unusual or isolated traffic patterns through random tree partitioning. This produced a second anomaly-related signal for each network traffic sample. Using this model was beneficial as it complemented the Autoencoder by evaluating abnormal events from a different perspective.

The next stage involved combining the two models. At this point, the outputs from the Autoencoder and Isolation Forest were merged into a new feature representation. This hybrid input was then used to train and apply the Logistic Regression meta-classifier. This step aimed to create a final system for predicting anomalies that could merge the strengths of both models. This stage is crucial because it turned two separate anomaly detectors into a more effective and coordinated prediction framework.

The testing phase then assessed the three main setups: the Autoencoder alone, the Isolation Forest alone and the final Hybrid model. By testing them separately and then comparing them, the study verified how each stage of implementation contributed to the outcome. This step was significant because it showed that the hybrid framework was not just a random mix of models, but a thoughtful and meaningful structure for predicting zero-day anomalies. It is also essential to note that the implementation was designed for future use. Although the current work used prepared traffic data, the framework can later support real-time integration. The steps for data input, anomaly scoring, hybrid decision-making, and result visualization can all be adapted for live monitoring. This makes the implementation suitable for academic evaluation and relevant for future practical applications in cybersecurity.

The overview shows that the testing and implementation process was systematic and thorough. It included data preparation, individual model development, hybrid integration and formal evaluation. As a result, this component successfully created a working anomaly detection framework for predicting zero-day attacks through network traffic patterns, while also supporting the goals of a larger real-time cyber threat prediction system.



6. Conclusion

This research project focused on predicting zero-day attacks using network traffic patterns as part of a larger Real-Time Intelligent System for Predicting Cyber Threats. The study aimed to tackle the growing challenge of detecting zero-day threats, which are particularly dangerous because they exploit unknown vulnerabilities and are hard to identify with traditional signature-based intrusion detection systems. As modern cyber threats evolve, it has become essential to develop security methods that go beyond reactive detection and enable more intelligent, proactive threat prediction. To address this issue, this research used an unsupervised approach to detect anomalies based on network traffic behavior. This method was chosen because zero-day attacks usually lack labeled datasets beforehand, making supervised learning less practical. Instead of depending on known attack signatures or fully labeled examples, the proposed framework learned the features of normal traffic and then spotted changes that might suggest suspicious or malicious activity. This made the method better suited for detecting new threats and more aligned with the reality of zero-day attack prediction.

The implementation used the HIKARI-2021 dataset because it features modern, realistic network traffic patterns, varied traffic behavior, and relevant intrusion scenarios. This dataset provided a suitable experimental base for developing anomaly detection models that fit today's cybersecurity environments better than older benchmark datasets. Thus, the choice of dataset was a major strength of the study, enhancing the real-world relevance of the system created.

The proposed framework was built using a hybrid design that included an Autoencoder, an Isolation Forest and a Logistic Regression meta-classifier. The Autoencoder learned normal network traffic behavior through reconstruction and generated anomaly signals based on reconstruction errors. The Isolation Forest contributed an additional anomaly signal by isolating unusual traffic points. The outputs from these two models were then combined in the meta-classifier, which produced the final prediction for anomalies. This hybrid structure allowed the framework to leverage the strengths of different unsupervised methods, creating a more effective prediction mechanism than a single-model approach. The research showed that analyzing anomalies in network traffic is an effective strategy for predicting zero-day attacks. Since network traffic frequently contains early signs of abnormal system behavior, it can be an important source for identifying suspicious activities before an attack is fully recognized or classified. By focusing on traffic anomalies instead of predefined attack rules, this component helps create a more adaptable model for cybersecurity defense.

Another significant contribution of this research is its integration within the broader cyber threat prediction system. This component does not work alone. Instead, it is a vital predictive layer within a larger framework that also considers system calls, user behavior, and endpoint log patterns. In this way, this research supports a more complete and intelligent cybersecurity architecture, where different data sources can work together to strengthen threat prediction. In conclusion, this study successfully developed and implemented a structured anomaly detection framework for predicting

zero-day attacks using network traffic patterns. It filled important gaps in previous work by using unsupervised learning, a modern dataset and a hybrid model tailored to the challenge of previously unseen cyber threats. Consequently, the component contributes significantly to the overall goal of creating an intelligent real-time system for predicting cyber threats. It also lays a solid foundation for future improvements, including live deployment, multi-source integration and further refining anomaly-based defense mechanisms.

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